Causal Separation in Portfolio Choice: Screening-Off Information and Conditional Risk

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Abstract

Conditional portfolio choice depends on the information used to define conditional moments, yet that information is typically taken as given. We introduce causal separation for portfolio choice: a portfolio-information principle in which horizon-closed conditioning screens asset returns into mutually conditionally independent components, with a common-cause structural model providing its causal interpretation. Exact separation yields a decision-time decomposition of risk into diagonal residual risk and uncertainty in horizon-closed conditional means. Projecting systematic risk onto finite-dimensional driver innovations produces a low-rank represented component, becoming exactly diagonal-plus-low-rank under affine response. The selected information also induces state-sensitivity constraints in a classical equality-constrained Markowitz problem. We derive exact finite covariance-perturbation identities and non-asymptotic bounds linking representation error to portfolio weights and the efficient frontier. Controlled experiments recover the theoretical decomposition, perturbation identities, and intervention behavior. Empirical applications to U.S. equities show substantial reductions in residual cross-sectional covariance and competitive out-of-sample structured covariance estimates in shorter training windows. Causal separation therefore makes the conditioning representation an explicit, testable component of portfolio construction.

Keywords: portfolio optimization; conditioning information; causal separation; screening-off; covariance estimation; mean–variance
JEL Classification: C58; C61; G11

1 Introduction

Conditional information has long been part of portfolio and asset-pricing theory. Intertemporal and conditional models show how state variables alter investment opportunities and conditional moments (Merton, 1973; Hansen and Richard, 1987). Portfolio research has developed efficient ways to use conditioning variables, construct state-dependent or mimicking portfolios, and incorporate manager information (Ferson and Siegel, 2001; Ferson et al., 2006; Chiang, 2015; Peñaranda and Sentana, 2016). More recently, high-dimensional conditioning sets have been handled through variable selection and dimension reduction (Pereira, 2021). In parallel, factor models, covariance shrinkage and portfolio constraints address instability once a risk representation has been specified (Fan et al., 2008; Ledoit and Wolf, 2004; Jagannathan and Ma, 2003).

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These lines of work solve different parts of the portfolio problem. Predictive selection asks which variables improve forecasts or portfolio performance, while covariance regularization asks how to estimate and stabilize risk after a representation has been chosen. What remains less developed is a criterion for choosing observable conditioning information from the cross-sectional dependence that remains after conditioning. This distinction matters because two representations can have similar predictive content while leaving very different residual dependence, and that residual dependence enters the covariance matrix optimized by the portfolio.

This paper introduces causal separation for portfolio choice. Its probabilistic component is screening-off: a representation separates the asset cross-section when horizon-closed conditioning information makes horizon returns mutually conditionally independent. We distinguish that information from the information available when the portfolio is chosen, so the separation property can be stated on realized historical windows without giving the portfolio access to future information. Under the common-cause structural conditions developed below, the separating information has a causal interpretation; observational screening alone does not identify causal labels.

The first consequence is an exact conditional-risk decomposition. Screening makes the residual conditional covariance diagonal, while the remaining systematic risk is the covariance of horizon-closed conditional means viewed from the decision date. Screening alone does not imply a low-rank covariance. A finite-dimensional projection of those conditional means onto observed driver innovations produces a low-rank represented component plus a positive-semidefinite response residual, and the low-rank form becomes exact under affine response. This distinction separates incomplete representation of systematic response from incomplete screening of residual cross-asset covariance.

The selected information also enters portfolio choice through decision-time state sensitivities. Neutralizing economically chosen state directions gives linear portfolio restrictions, after which the optimization is the standard equality-constrained Markowitz problem (Markowitz, 1952). The screening step determines the conditional risk representation and admissible directions; the perturbation results then quantify how remaining covariance error changes optimal weights and the efficient frontier.

The paper makes four main contributions. First, it derives the exact conditional-risk decomposition implied by screening without requiring Gaussianity or a factor model. Second, it characterizes the finite-dimensional innovation representation and the additional condition under which diagonal-plus-low-rank covariance is exact. Third, it formalizes causal separation for portfolio choice by showing that a common-cause structural partition is sufficient for exact portfolio screening, while observationally equivalent separators need not have the same causal meaning. Fourth, it derives exact finite covariance-perturbation identities and non-asymptotic bounds that connect residual screening error and omitted response risk to portfolio sensitivity.

The evidence is organized in two layers. Controlled experiments evaluate the theoretical results in settings where the population structure is known: they recover the distinction between decision and horizon-closed information, verify the covariance and perturbation identities, isolate response-representation and screening errors, and show how an observationally equivalent proxy separates from the common-state representation under intervention. The empirical applications then examine how the same construction behaves with market data. In the empirical screening study of Section 8.2, selected driver representations materially reduce residual covariance relative to unconditional risk and matched-dimension principal components, while the broad-market augmentation reveals how the selected information set changes when the candidate universe is enriched. In the point-in-time covariance study of Section 8.3, the structured covariance is competitive with established factor and shrinkage estimators, particularly in the shorter estimation window. Together, these experiments connect the theoretical objects to observable residual dependence, covariance estimation and portfolio risk.

The resulting causal-separation framework turns the conditioning representation from a fixed input of portfolio optimization into an explicit object of analysis. Screening-off provides the probabilistic principle for selecting and assessing that representation, determines the associated conditional-risk decomposition, and distinguishes represented systematic variation from residual cross-asset dependence. The portfolio problem that follows remains classical in form, but its information structure is now explicit and the consequences of imperfect representation can be propagated exactly to optimal weights and the efficient frontier. This creates a direct link from information selection to conditional risk and portfolio choice.

2 Related literature

The literature can be read as a sequence of increasingly explicit choices about information and risk. In dynamic asset-pricing models, conditioning variables determine investment opportunities, conditional moments and pricing restrictions (Merton, 1973; Hansen and Richard, 1987; Hansen and Jagannathan, 1991). Portfolio research then brought conditioning information directly into portfolio construction. Ferson and Siegel (2001) derive minimum-variance efficient portfolios that exploit conditioning information, Ferson et al. (2006) characterize mimicking portfolios built from predetermined state variables, Chiang (2015) studies benchmark-relative portfolio choice when managers possess conditioning information, and Peñaranda and Sentana (2016) analyze mean–variance frontiers with conditioning information. This literature establishes why conditioning matters for portfolio choice, but the conditioning representation is typically supplied as part of the model or as a set of predictor variables.

As the number of candidate variables grows, the problem becomes one of statistical information selection. Pereira (2021) studies high-dimensional conditioning information using variable selection, shrinkage, principal-component regression and partial least squares evaluated by out-of-sample portfolio criteria. More broadly, the sensitivity of mean–variance portfolios to estimated moments is well known (Michaud, 1989; Best and Grauer, 1991; Kan and Zhou, 2007), and parameter or model uncertainty has motivated robust and multi-prior portfolio methods (Garlappi et al., 2007). These approaches ask which variables or models improve prediction, estimation or portfolio performance. The criterion developed here is different: the selected information is assessed by the cross-asset dependence that remains after conditioning. Predictive relevance and screening are therefore related but distinct properties.

A parallel literature addresses the risk-estimation problem once a representation has been specified. Classical and high-dimensional factor models reduce covariance dimension (Chamberlain and Rothschild, 1983; Fan et al., 2008), while characteristic-based latent-factor models provide another conditional representation of the cross-section (Kelly et al., 2019). Covariance shrinkage stabilizes estimation without requiring an explicit economic factor representation (Ledoit and Wolf, 2004), portfolio constraints can themselves regularize weights (Jagannathan and Ma, 2003), and improvements in covariance estimation need not translate mechanically into portfolio gains (DeMiguel et al., 2009). Recent work extends this line through nonlinear covariance filtering (Bongiorno and Challet, 2024), portfolio-specific evaluation of covariance estimators (Dom et al., 2025), mixed-frequency factor information (Lu and Wang, 2025), and dynamic hierarchical covariance estimation (Wang et al., 2026). These methods regularize or estimate a covariance object conditional on a chosen representation. Screening addresses an earlier question: what observable information should be conditioned on, and what covariance structure follows when that information removes common dependence?

This distinction also clarifies the relation to factor models. Under affine response to observed innovations, screening produces a diagonal-plus-low-rank covariance of familiar factor form. Low rank, however, is not implied by screening itself; it follows from the additional finite-dimensional response representation. The observed innovation coordinates also need not coincide with principal components. The matched-dimension PCA comparisons in the empirical analysis therefore serve as a natural statistical benchmark rather than as a competing definition of the information set. In this sense, screening complements rather than replaces factor modeling or shrinkage: it specifies an information-conditioned decomposition on which those estimation methods can subsequently operate.

The final connection is to the probabilistic and causal literature on screening-off. The common-cause idea originates with Reichenbach (1956) and is formalized through conditional-independence structures in graphical models (Pearl, 1988, 2009). Modern causal inference distinguishes observational conditional independence from invariance under intervention (Peters et al., 2016, 2017), while general conditional-independence testing is known to be statistically difficult (Shah and Peters, 2020). Causal portfolio formulations have begun to give economic drivers a structural interpretation (Rodríguez Domínguez, 2023, 2024, 2025a,b). The contribution here is not the general notion of screening-off, which has a much longer history, but its portfolio-specific combination with a common-cause structural representation and the conditional-risk consequences derived from that separation. We call this construction *causal separation for portfolio choice*. Screening-off defines the probabilistic separation criterion; the common-cause model supplies the causal interpretation. Observationally equivalent separators can therefore generate the same conditional portfolio objects without identifying the same causal variables.

3 Screening-off and portfolio information

3.1 Decision information and horizon-closed information

Let $\mathcal{A} = \{1, \ldots, n\}$ be the asset universe and let $X = (X^1, \ldots, X^K)$ be a finite declared universe of observable candidate drivers. Fix a decision date t, lookback $\delta > 0$, and horizon $h \in (0, H]$. For $\mathcal{D} \subseteq \{1, \ldots, K\}$ define

$$\mathcal{F}^{\mathcal{D}}_{t,0} := \sigma(X^j_s : j \in \mathcal{D},\ s \in [t - \delta, t]), \qquad \mathcal{F}^{\mathcal{D}}_{t,h} := \sigma(X^j_s : j \in \mathcal{D},\ s \in [t - \delta, t + h]). \tag{1}$$

Let $\mathcal{F}^{\mathcal{A}}_t$ denote asset information available at the decision date. The two nested information sets are

$$\mathcal{G}_t(\mathcal{D}) := \mathcal{F}^{\mathcal{A}}_t \vee \mathcal{F}^{\mathcal{D}}_{t,0}, \qquad \mathcal{H}_{t,h}(\mathcal{D}) := \mathcal{F}^{\mathcal{A}}_t \vee \mathcal{F}^{\mathcal{D}}_{t,h}. \tag{2}$$

Portfolio weights are $\mathcal{G}_t(\mathcal{D})$-measurable. The larger $\mathcal{H}_{t,h}(\mathcal{D})$ is a conditioning object used to express screening after the common horizon innovations have realized; it is not information available when the portfolio is chosen.

Definition 3.1 (Exact portfolio separator)**.** A candidate set $\mathcal{D}$ is an *exact portfolio separator* at t if, for every $h \in (0, H]$,

$$\mathcal{L}(r_{t+h} \mid \mathcal{H}_{t,h}(\mathcal{D})) = \bigotimes_{i=1}^{n} \mathcal{L}(r^i_{t+h} \mid \mathcal{H}_{t,h}(\mathcal{D})) \quad \text{a.s.} \tag{3}$$

The definition requires mutual, not merely pairwise, conditional independence. It also explains why the two information windows cannot generally be collapsed. If

$$r^i_{t+h} = b_i \eta_{t,h} + \varepsilon^i_{t,h}, \tag{4}$$

with a common innovation $\eta_{t,h}$ realized after t, then conditioning only on decision information leaves the cross-sectional covariance generated by $\eta_{t,h}$. Historical horizon-closed regression can condition on realized $(\eta_{s,h}, r_{s+h})$ pairs for $s < t$ without making w_t depend on future information.

Screening and predictive sufficiency are different requirements. Relative to the declared full universe, a selected set can be called sufficient if replacing the full driver information by the selected information leaves the relevant decision-time and horizon-closed conditional laws unchanged. Because the declared universe is finite, inclusion-minimal sufficient screening sets exist whenever the sufficient class is nonempty, but named-variable representations need not be unique. Two sets that generate the same windowed sigma-algebras are observationally equivalent for all conditional-law functionals used below.

3.2 Causal interpretation under additional structure

The portfolio theory requires Definition 3.1, not a causal graph. A causal interpretation is available under stronger assumptions and only in one direction.

Assumption 3.2 (Common-cause structural partition)**.** For every relevant horizon, the variables admit an acyclic modular structural causal model with mutually independent exogenous blocks $U^0, U^1, \ldots, U^n$ such that: (i) a common state path $C_{t,h} = c_h(U^0)$ is represented by a declared set $\mathcal{D}_C$ on both the decision and horizon-closed windows; (ii) each asset path satisfies $R^i_{[t-\delta,t+h]} = f_{i,h}(C_{t,h}, U^i)$; and (iii) all common causal channels among the assets over the horizon are contained in $C_{t,h}$.

Definition 3.3 (Causal separator for portfolio choice)**.** Under Assumption 3.2, the declared representation $\mathcal{D}_C$ that generates the common-state information is called a *causal separator* for portfolio choice. We refer to the associated construction as *causal separation*.

Proposition 3.4 (Causal soundness of screening)**.** *Under Assumption 3.2, the representation $\mathcal{D}_C$ generating exactly the common-state information is an exact portfolio separator. A strictly richer conditioning set requires its own product-law or collider-safety check.*

The proof is given in Appendix A. Thus causal separation implies exact portfolio separation under Assumption 3.2; the converse is not claimed. The restriction to the information generated by the common state is substantive: conditioning on additional variables can open dependence paths even when the common-cause state itself screens the assets.

Proposition 3.5 (Observational equivalence and intervention)**.** *Suppose an exact separator C and an observable representation S generate the same decision and horizon-closed information. Then they generate identical conditional laws and identical portfolio objects. Observational screening therefore identifies operative information, not unique causal labels. Under Assumption 3.2, an intervention on a variable outside the ancestral system of the common state and horizon returns leaves the common-state conditional laws invariant, provided the intervention does not alter the conditioning information itself.*

An invertible recording $P = \psi(C)$ is therefore indistinguishable from C observationally, even if it is only a measurement of the underlying state. If the measurement mechanism is later severed from C, the proxy can lose screening while the common state remains valid. This is the distinction tested in the controlled intervention experiment.

4 Conditional risk under screening-off

Fix a representation and horizon and write $\mathcal{G}_t := \mathcal{G}_t(\mathcal{D})$ and $\mathcal{H}_{t,h} := \mathcal{H}_{t,h}(\mathcal{D})$. Assume r_{t+h} is square-integrable. Define

$$\mu_{t,h} := \mathbb{E}[r_{t+h} \mid \mathcal{G}_t], \qquad m_{t,h} := \mathbb{E}[r_{t+h} \mid \mathcal{H}_{t,h}]. \tag{5}$$

4.1 Exact decomposition

Theorem 4.1 (Decision-time risk decomposition)**.** *If $\mathcal{D}$ is an exact portfolio separator, then*

$$Q_{t,h} := \mathrm{Cov}(r_{t+h} \mid \mathcal{G}_t) = D_{t,h} + \Sigma^{\mathrm{sys}}_{t,h}, \tag{6}$$

where

$$D_{t,h} := \mathbb{E}[\mathrm{Cov}(r_{t+h} \mid \mathcal{H}_{t,h}) \mid \mathcal{G}_t] \tag{7}$$

is diagonal positive semidefinite and

$$\Sigma^{\mathrm{sys}}_{t,h} := \mathrm{Cov}(m_{t,h} \mid \mathcal{G}_t) \succeq 0. \tag{8}$$

The theorem follows from the conditional law of total covariance together with screening and is the main structural implication. It does *not* imply low rank. With scalar $Z \sim N(0,1)$ and independent idiosyncratic errors, the three conditional means Z, $Z^2 - 1$ and $Z^3 - 3Z$ are mutually uncorrelated and generate a rank-three systematic covariance even though the separator is one-dimensional.

4.2 Finite-dimensional innovation representation

Choose a square-integrable horizon innovation vector $\xi_{t,h} \in \mathbb{R}^k$, measurable with respect to $\mathcal{H}_{t,h}$, and let

$$\widetilde{\xi}_{t,h} := \xi_{t,h} - \mathbb{E}[\xi_{t,h} \mid \mathcal{G}_t], \qquad \Lambda_{t,h} := \mathrm{Cov}(\xi_{t,h} \mid \mathcal{G}_t) \succ 0. \tag{9}$$

Theorem 4.2 (Projection of systematic risk)**.** *Define the conditional linear projection*

$$L_{t,h} := \mathrm{Cov}(m_{t,h}, \xi_{t,h} \mid \mathcal{G}_t)\Lambda^{-1}_{t,h}, \qquad u_{t,h} := m_{t,h} - \mu_{t,h} - L_{t,h}\widetilde{\xi}_{t,h}. \tag{10}$$

Then $\mathrm{Cov}(u_{t,h}, \widetilde{\xi}_{t,h} \mid \mathcal{G}_t) = 0$ *and*

$$\Sigma^{\mathrm{sys}}_{t,h} = L_{t,h}\Lambda_{t,h}L^{\top}_{t,h} + \Omega_{t,h}, \qquad \Omega_{t,h} := \mathrm{Cov}(u_{t,h} \mid \mathcal{G}_t) \succeq 0, \tag{11}$$

and therefore, under exact screening,

$$Q_{t,h} = D_{t,h} + L_{t,h}\Lambda_{t,h}L^{\top}_{t,h} + \Omega_{t,h}. \tag{12}$$

Equation (12) separates the exact screening implication from the additional response representation. If the centered horizon-closed conditional mean is affine in $\widetilde{\xi}_{t,h}$, then $\Omega_{t,h} = 0$ and

$$Q_{t,h} = D_{t,h} + L_{t,h}\Lambda_{t,h}L^{\top}_{t,h}, \qquad \mathrm{rank}(Q_{t,h} - D_{t,h}) \le k. \tag{13}$$

For a smooth nonlinear response, a small-innovation expansion gives a local low-rank approximation rather than an exact identity. Corollary A.1 and Proposition A.2 in Appendix A state the exact affine case and the local nonlinear case under conditional L^2 regularity.

4.3 Decision-time sensitivity geometry

The innovation loading $L_{t,h}$ describes uncertainty arriving over the horizon. Portfolio sensitivity restrictions use a different object: the response of the decision-time conditional mean to the state already observed at t. Suppose

$$\mu_{t,h} = g_t(Z_t), \qquad Z_t \in \mathbb{R}^m, \tag{14}$$

with g_t continuously differentiable locally. Define

$$S_t := Dg_t(Z_t) \in \mathbb{R}^{n\times m}. \tag{15}$$

If $U_t \in \mathbb{R}^{m\times q}$ spans economically chosen state directions to be neutralized, then

$$w^\top S_t U_t = 0 \iff \Gamma_t w = 0, \qquad \Gamma_t := U_t^\top S_t^\top. \tag{16}$$

The choice of U_t is an admissibility decision, not an output of screening. Under a local diffeomorphism $z' = \psi(z)$, transporting directions as $U_t' = D\psi(Z_t)U_t$ and sensitivities as $S_t' = S_t D\psi(Z_t)^{-1}$ gives $U_t'^\top S_t'^\top = U_t^\top S_t^\top = \Gamma_t$. Hence the portfolio restriction depends on the economic directions rather than on the chosen coordinates. Proposition A.3 in Appendix A records the invariance formally.

5 Projected conditional mean–variance choice

Once (μ, Q) and the admissibility geometry have been specified, the optimization is the classical mean–variance problem (Markowitz, 1952). At a fixed decision node write $\Gamma := \Gamma_t$. For risk tolerance $\gamma \geq 0$, stack the budget and sensitivity restrictions as

$$A := \begin{pmatrix} \mathbf{1}^\top \\ \Gamma \end{pmatrix}, \qquad b := (1, 0, \ldots, 0)^\top, \tag{17}$$

with A full row rank, and consider

$$\max_w \left\{ \gamma w^\top \mu - \frac{1}{2} w^\top Q w \right\} \quad \text{subject to} \quad Aw = b, \qquad Q \succ 0. \tag{18}$$

Theorem 5.1 (Projected conditional Markowitz portfolio)**.** *The unique solution of* (18) *is*

$$w^\star = w_0 + \gamma M \mu, \tag{19}$$

where

$$w_0 := Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} b, \tag{20}$$
$$M := Q^{-1} - Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} A Q^{-1}. \tag{21}$$

If N spans $\ker A$*, then*

$$M = N (N^\top Q N)^{-1} N^\top, \tag{22}$$

so M is the inverse risk operator restricted to feasible directions.

Define

$$\mu_0 := w_0^\top \mu, \qquad \sigma_0^2 := w_0^\top Q w_0, \qquad \Delta := \mu^\top M \mu \geq 0. \tag{23}$$

Then

$$\mu_p(\gamma) = \mu_0 + \gamma\Delta, \qquad \sigma_p^2(\gamma) = \sigma_0^2 + \gamma^2 \Delta. \tag{24}$$

If $\Delta > 0$, eliminating γ gives

$$\sigma_p^2 = \sigma_0^2 + \frac{(\mu_p - \mu_0)^2}{\Delta}. \tag{25}$$

Moreover,

$$\sqrt{\Delta} = \sup_{v \in \ker A,\, v \neq 0} \frac{|v^\top \mu|}{\sqrt{v^\top Q v}}, \tag{26}$$

so $\sqrt{\Delta}$ is the maximum conditional information ratio of a zero-cost feasible perturbation. It is not, in general, the maximum Sharpe ratio among fully invested portfolios.

The diagonal residual risk is also useful for conditioning. If

$$D \succeq \underline{\sigma}^2 I, \tag{27}$$

then $Q \succeq \underline{\sigma}^2 I$ and $\|M\|_2 \leq \underline{\sigma}^{-2}$. Here and below, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the spectral norm for matrices. Consequently,

$$\|w^\star(\mu + \delta\mu) - w^\star(\mu)\|_2 \leq \frac{\gamma}{\underline{\sigma}^2} \|\delta\mu\|_2. \tag{28}$$

Screening identifies the residual block; the positive variance floor is an additional quantitative assumption. Proposition A.4 in Appendix A gives the corresponding operator argument.

When the affine covariance representation (13) is used and the positive residual floor (27) holds, D is invertible and Woodbury gives

$$Q^{-1}x = D^{-1}x - D^{-1}L(\Lambda^{-1} + L^\top D^{-1} L)^{-1} L^\top D^{-1} x. \tag{29}$$

For fixed driver and constraint dimensions this reduces the dominant computational cost to linear order in the number of assets; Proposition A.5 in Appendix A gives the corresponding arithmetic and memory bounds. If $\Omega \neq 0$ and no additional structure is available, the formula is an approximation rather than the inverse of the true covariance.

6 Misspecification and robustness

The exact separator and affine-response cases are benchmarks. Empirical implementations can fail in two different ways. The chosen finite innovation span can miss systematic response variation even when screening is exact, and the selected information can leave cross-asset residual covariance even when the systematic response is well represented.

6.1 Two covariance-error channels

For any candidate representation, the law of total covariance gives

$$Q = V + \mathrm{Cov}(m \mid \mathcal{G}_t), \qquad V := \mathbb{E}[\mathrm{Cov}(r \mid \mathcal{H}_{t,h}) \mid \mathcal{G}_t]. \tag{30}$$

Let $D := \mathrm{diag}(V)$ denote the diagonal matrix with the same diagonal as V and define

$$E := V - D. \tag{31}$$

Then E is symmetric with zero diagonal. Applying Theorem 4.2 to the systematic term gives the exact decomposition

$$\boxed{Q = D + L\Lambda L^\top + \Omega + E.} \tag{32}$$

The two error channels are therefore

$$R := \Omega + E, \qquad Q^0 := D + L\Lambda L^\top, \qquad Q = Q^0 + R. \tag{33}$$

A nonzero $\Omega \succeq 0$ is compatible with exact screening; it says that the chosen innovation representation is incomplete or too linear. A nonzero E means that second-moment screening is incomplete. Exact screening implies $E = 0$, but $E = 0$ does not imply the product law outside distributional classes in which second moments characterize dependence.

6.2 Exact finite covariance perturbations

For this subsection, let $\bar{Q} \succ 0$ denote any baseline covariance and let $\bar{Q}_R := \bar{Q} + R \succ 0$, holding (μ, A, b, γ) fixed. Let $(M, w^\star, \Delta)$ be the projected objects under $\bar{Q}$ and $(M_R, w_R^\star, \Delta_R)$ their counterparts under $\bar{Q}_R$. This generic perturbation notation is distinct from the decomposition $Q = Q^0 + R$ in (33).

Theorem 6.1 (Finite covariance-perturbation identities)**.** *Whenever both covariance matrices are positive definite,*

$$M_R - M = -MRM_R = -M_R RM, \tag{34}$$
$$w_R^\star - w^\star = -M_R R w^\star = -MRw_R^\star, \tag{35}$$
$$\Delta_R - \Delta = -(M\mu)^\top R(M_R\mu). \tag{36}$$

These are finite identities, not first-order approximations. For $\bar{Q}_\tau = \bar{Q} + \tau R$ they imply

$$w_\tau^\star - w^\star = -\tau MRw^\star + O(\tau^2), \tag{37}$$
$$\Delta_\tau - \Delta = -\tau(M\mu)^\top R(M\mu) + O(\tau^2). \tag{38}$$

If the baseline is the structured covariance $\bar{Q} = Q^0 = D + L\Lambda L^\top$, the floor (27) holds, and $\|R\|_2 < \underline{\sigma}^2$, then

$$\|w_R^\star - w^\star\|_2 \le \frac{\|R\|_2}{\underline{\sigma}^2 - \|R\|_2}\|w^\star\|_2, \tag{39}$$

and

$$|\Delta_R - \Delta| \le \frac{\|R\|_2}{\underline{\sigma}^2(\underline{\sigma}^2 - \|R\|_2)}\|\mu\|_2^2. \tag{40}$$

The sign structure also matters. If $Q_{\text{true}} = Q^0 + \Omega$ with $\Omega \succeq 0$, and (M_0, Δ_0) and $(M_{\text{true}}, \Delta_{\text{true}})$ are computed under Q^0 and Q_{true}, respectively, then

$$M_{\text{true}} \preceq M_0, \qquad \Delta_{\text{true}} \le \Delta_0. \tag{41}$$

Omitting systematic response covariance can therefore only overstate feasible frontier potential. No analogous sign statement holds for the indefinite residual-screening error E.

6.3 Residual covariance diagnostic

Mean–variance choice depends directly on second moments, so the empirical implementation uses a weaker second-moment diagnostic than the full product-law definition. Assume $V_{ii} > 0$, put $s_i^2 := V_{ii}$ and define

$$\epsilon^{\text{cov}}(\mathcal{D}) := \max_{i<j} \frac{|E_{ij}|}{s_i s_j}. \tag{42}$$

An exact separator implies $\epsilon^{\text{cov}} = 0$, not conversely.

Proposition 6.2 (Diagnostic-to-operator bound)**.** *If* $|E_{ij}| \le \epsilon^{\text{cov}} s_i s_j$ *for all* $i \neq j$*, then*

$$\|E\|_2 \le \|E\|_F \le \epsilon^{\text{cov}} \sqrt{\left(\sum_i s_i^2\right)^2 - \sum_i s_i^4} \le \epsilon^{\text{cov}} \|s\|_2^2. \tag{43}$$

Combining (43) with the perturbation bounds connects a pairwise residual-correlation diagnostic to portfolio sensitivity. It does not transform the diagnostic into a test of mutual conditional independence. A training diagnostic can be used for representation selection; on a disjoint segment we distinguish a *set-refit* diagnostic, which fixes the selected labels and re-estimates nuisance models, from a *frozen-model* diagnostic, which carries the fitted parameters forward. Neither is automatically a statistical upper confidence bound. The term *certified* is reserved for a procedure with an explicit population coverage guarantee. Corollary A.6 in Appendix A gives the simultaneous first-order effect of mean and covariance perturbations.

7 Controlled evidence

The controlled experiments use data-generating processes for which the population structure is known. Their role is to separate the theoretical implications and to check the numerical implementation; the proofs do not rely on simulation evidence. Unless stated otherwise, driver states follow independent Gaussian AR(1) processes with coefficient 0.95, with innovation variances scaled so that the stationary driver variances equal one. Asset loadings are drawn from centered Gaussian distributions and idiosyncratic daily volatilities lie between 0.8% and 2%.

7.1 Two information windows

The first experiment isolates the distinction between decision information and horizon-closed screening information. A Gaussian panel with $n = 30$ assets and $m = 3$ common drivers is residualized first on the current state alone and then on the state together with the realized driver innovation. The maximal absolute off-diagonal residual correlation is 0.83 under past-only conditioning and 0.025 under horizon-closed conditioning on a long simulated path. Figure 1 displays the two residual-correlation matrices.

The experiment illustrates the covariance implication of screening rather than estimating the full product law from data. In this Gaussian design the stronger independence statement is known from the construction.

7.2 Out-of-sample covariance estimation

The second experiment asks whether exploiting an observed affine innovation structure helps when that structure is correctly specified. Global-minimum-variance portfolios are formed from the sample covariance, Ledoit–Wolf shrinkage, PCA with different factor counts, the structured estimator $\widehat{D} + \widehat{L}\widehat{\Lambda}\widehat{L}^\top$, and the oracle covariance. In each training sample, driver AR dynamics are fitted with intercepts, historical innovations are extracted, returns are regressed on lagged driver states and those innovations, and the covariance is assembled from the estimated innovation covariance and residual variances. The design uses $n \in \{50, 100, 200\}$, training lengths $T \in \{126, 252, 504\}$, 250 Monte Carlo replications per cell and 252 out-of-sample observations. Figure 2 reports the $T = 252$ slice and its variance calibration.

The unrestricted sample estimator deteriorates as n/T rises. The structured estimator remains close to the oracle in realized volatility and variance calibration, while PCA supplied with the correct

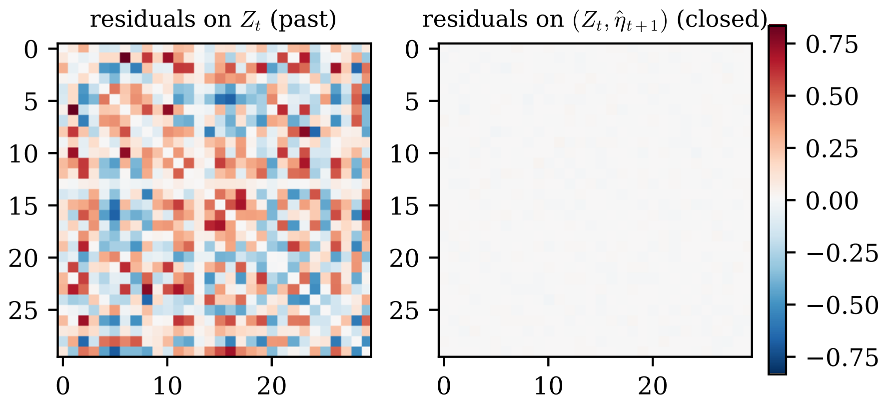

Figure 1: Controlled two-window illustration. Conditioning only on the decision-time state leaves the common horizon innovation in the residual covariance (left). Conditioning on the horizon-closed state and realized innovation removes that systematic block in the Gaussian normal form (right). The portfolio decision itself uses only date-t information.

factor dimension is also close. The comparison is therefore not evidence that observable-driver structure must beat a correctly dimensioned latent factor model. It shows that, when the affine screening representation is correct, the theoretical covariance form can be estimated without first recovering latent factors. Ledoit–Wolf remains a strong benchmark and is not uniformly dominated. The full experiment grid is reported in Appendix B.

7.3 Separating response error from screening error

The robustness experiment uses the decomposition $R = \Omega + E$ directly. The baseline has $n = 60$ assets and $m = 3$ Gaussian driver innovations. Let $\underline{\sigma}^2 := \min_i D_{ii}$ and parameterize perturbation size by

$$\rho := \frac{\|R\|_2}{\underline{\sigma}^2} \in [0, 0.8]. \tag{44}$$

For the response channel, screening is exact and the horizon-closed conditional mean contains a quadratic Hermite component orthogonal to the linear innovation span, producing $\Omega_\rho \succeq 0$ and $E = 0$. For the screening channel, the systematic response is affine and a zero-diagonal residual covariance perturbation produces $E \neq 0$ and $\Omega = 0$. Figure 3 compares the exact portfolio displacement, its first-order approximation and the finite bound across the two channels.

Across the perturbation grid, the exact identities in Theorem 6.1 hold to numerical precision. At $\rho = 0.5$, omitted response risk moves the portfolio by 13.4% in relative Euclidean norm and reduces Δ by 0.39%; the first-order values are 14.4% and −0.42%. Reversing the sign of the residual-screening perturbation reverses the direction of its effect on Δ, illustrating why the one-sided result is specific to $\Omega \succeq 0$.

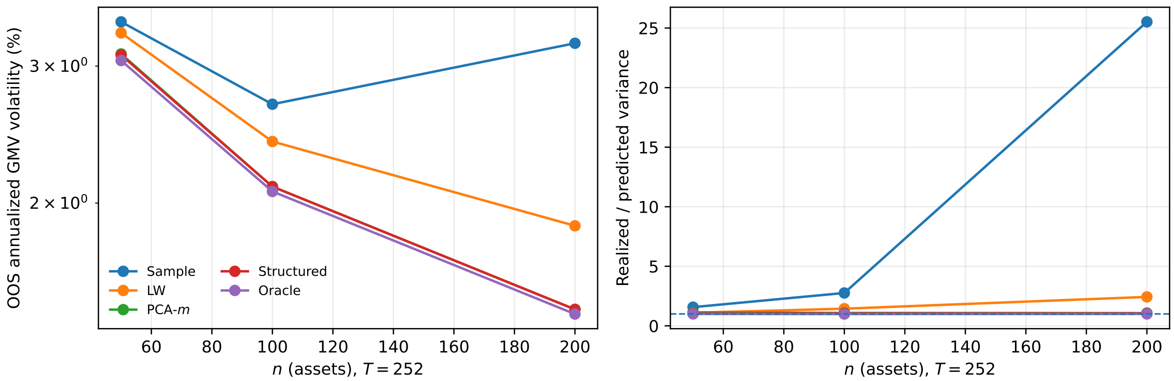

Figure 2: Controlled covariance estimation at $T = 252$. Left: out-of-sample annualized GMV volatility. Right: realized-to-predicted variance, where one denotes perfect calibration. The structured estimator is close to the oracle under the correctly specified affine design; PCA with the correct factor count provides an important latent-factor benchmark.

7.4 Observational equivalence broken by intervention

The final controlled experiment distinguishes observational equivalence from causal invariance. Returns are generated from a two-state normal form with $n = 40$. The common-state representation uses (Z^1, Z^2). The alternative records the first state through

$$P_t = -0.4 + 1.7 Z_t^1, \tag{45}$$

so $\sigma(P, Z^2) = \sigma(Z^1, Z^2)$ before intervention. With intercepts in both stages of estimation, the two representations produce fitted covariances, conditional means and portfolio weights that agree to numerical precision across 250 Monte Carlo replications.

At the intervention date the proxy measurement equation is replaced by

$$do\Big(P_t := -0.4 + 1.7 \widetilde{Z}_t^1\Big), \tag{46}$$

where $\widetilde{Z}^1$ is an independent stationary AR(1) copy with the same marginal process law as Z^1. The structural equations for the true state and returns are unchanged. Figure 4 shows that the two representations coincide before the intervention and separate afterwards.

Before intervention the mean holdout maximal absolute residual correlation is 0.212 for both representations. After intervention it is 0.211 for the common-state model and 0.911 for the proxy. Frictionless synthetic Sharpe ratios are identical before intervention and separate afterwards. These return statistics are model diagnostics, not claims about implementable performance. The identification result is the divergence in screening after the intervention, not the pre-intervention fit.

8 Empirical evidence

The empirical exercises are narrower than the controlled experiments. They measure residual covariance after conditioning and compare covariance estimators out of sample. They do not identify causal labels.

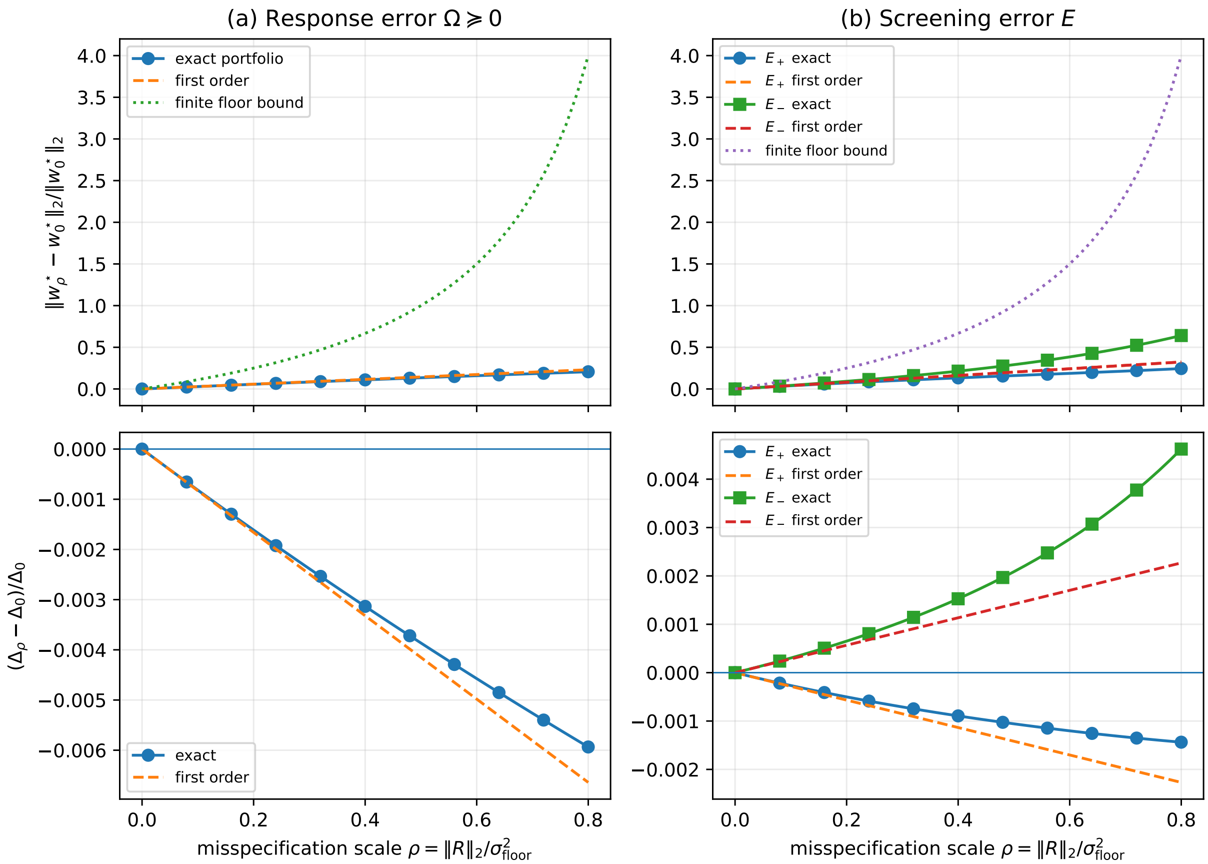

Figure 3: Two covariance-misspecification channels. Left: exact screening with nonlinear response and $\Omega_\rho \succeq 0$. Right: affine response with zero-diagonal residual screening error E_ρ of either sign. Top panels show relative portfolio displacement with the first-order approximation and finite bound; bottom panels show the relative change in frontier potential.

8.1 Data and point-in-time design

The first panel uses Bloomberg daily closing histories for 12 S&P 500 constituents and 127 candidate economic series. The raw universe spans developed- and emerging-market equity indices, sovereign curves and swap rates, currency crosses, commodity futures, credit aggregates and the VIX. Four series are excluded ex ante for frequency or data-integrity reasons. Rates and other series for which log changes are not well defined are first-differenced; the remaining positive price-like series are log-differenced. The resulting panel contains 123 transformed daily candidates and 5,653 aligned observations from 2 January 2001 to 30 June 2023. For any candidate set, driver dynamics are fitted with intercepts, historical innovations are extracted, and the closed-window return regression produces the residual correlations used in selection. On each 756-day rolling window, the training representation approximately minimizes $\widehat{\epsilon}^{\text{cov}}_{\text{train}}(\mathcal{D}) + \kappa|\mathcal{D}|$ with $\kappa = 0.01$ subject to $|\mathcal{D}| \leq 8$, using greedy forward selection followed by backward elimination. The first 60% of the window is used for selection, and the disjoint final 40% for set-refit or frozen-model evaluation. Appendix C lists the complete declared universes and gives the exact deterministic selection rule.

The second panel contains a fixed survivorship-conditioned list of 150 U.S.-listed equities from Bloomberg daily closing-price histories. Returns are reconstructed from raw prices rather than

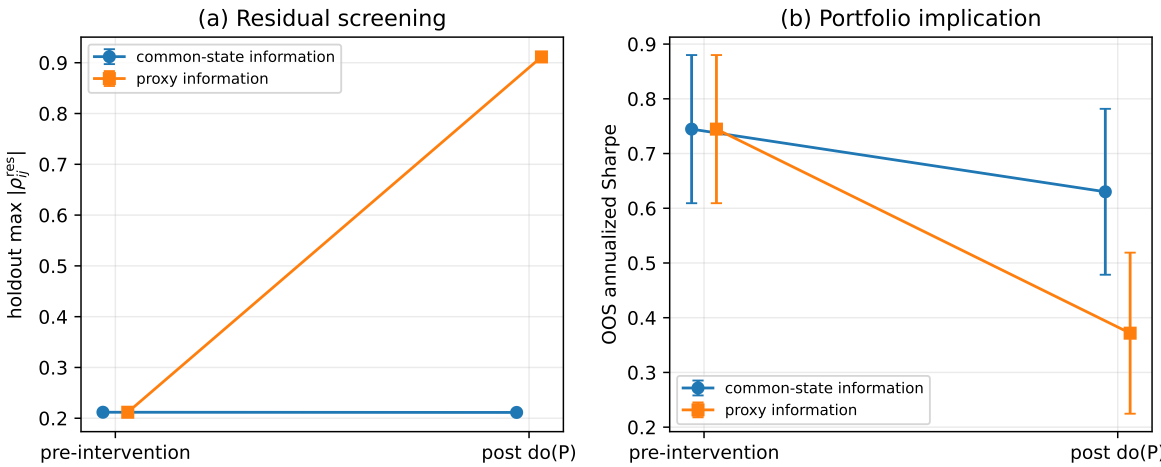

Figure 4: Information-equivalent proxy under intervention. Points are Monte Carlo means and error bars are ±2 Monte Carlo standard errors. Before intervention the common-state and proxy representations generate numerically identical portfolio objects. After the proxy measurement mechanism is severed from the state, residual screening deteriorates for the proxy while the common-state representation remains stable.

forward/backward-filled levels. Dates without actual observations for all names are removed. Split-like jumps are adjusted only when a large price ratio is within 5% of a predeclared conventional split factor between 1:20 and 20:1; other large moves are retained. This produces 3,267 daily return observations from 1 July 2010 to 30 June 2023. For every test fold, driver eligibility is determined only from representation selections completed before that fold begins; the representation is then re-selected from that point-in-time eligible pool and the covariance model is estimated using only the current training window. The design therefore prevents later representation selections from determining earlier covariance tests. Table 1 summarizes the two empirical designs, and Table 6 in Appendix C lists the fixed equity universe.

Table 1: Empirical designs. The first panel evaluates residual covariance after information selection; the second compares covariance estimators on a fixed equity panel.

Exercise	Assets and dates	Conditioning universe	Evaluation design
Screening	12 stocks, 2001–2023	123 transformed daily drivers	756-day windows; 60/40 selection/holdout split; set-refit and frozen-model diagnostics
Covariance	150 stocks, 2010–2023	Point-in-time eligible selected drivers	252 or 504 training days; 126-day non-overlapping test blocks; GMV comparison

8.2 Residual covariance and omitted-driver stress

Across 39 overlapping rolling windows, the frozen-model median maximal absolute residual correlation is 0.450 for selected driver sets and 0.868 for matched-dimension PCA. The corresponding frozen-model median mean-absolute residual correlations are 0.180 and 0.218, against 0.379 without conditioning. Selected sets have the lower mean-absolute diagnostic in 72% of windows. The set-refit

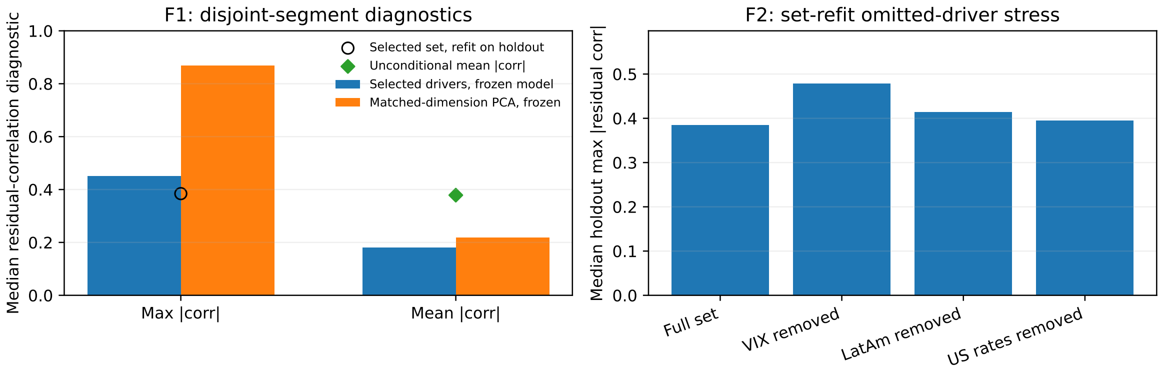

Figure 5: Empirical residual-covariance diagnostics. Left: frozen-model holdout diagnostics for selected driver representations and matched-dimension PCA; the open circle shows the selected-set diagnostic after refitting on the disjoint segment. Right: omitted-driver stress evaluated using the set-refit diagnostic. All quantities are second-moment diagnostics rather than causal tests or confidence bounds.

median maximal diagnostic is 0.384.

These results show that conditioning on selected drivers reduces residual covariance without eliminating it. The output is therefore described as a selected driver representation rather than an exact separator. The VIX appears in all 39 original selections, but because the rolling windows overlap, selection frequencies are descriptive persistence measures rather than independent-trial significance statements.

The omitted-driver stress uses the set-refit diagnostic. Removing the VIX raises the median maximal residual correlation from 0.384 to 0.478 and increases the diagnostic in 87% of windows. Removing the Latin-American equity block or the U.S.-rates block gives smaller medians of 0.415 and 0.395. A model-based omitted-channel floor computed from fitted loadings and residual innovation variance has correlation 0.56 with the realized VIX-removal increase. Appendix C gives the construction of this diagnostic. These are sensitivity results for the declared information universe, not causal-parent tests. Figure 5 summarizes the holdout diagnostics and the omitted-driver stress.

A sensitivity analysis of the candidate universe adds the CRSP value-weighted U.S. market return (including distributions, variable `vwretd`) as a broad market-information proxy to the compact 123-driver universe and repeats the same rolling selection. The median set-refit maximal residual correlation falls from 0.384 to 0.287, and the proxy is selected in 37 of 39 windows. The selection frequencies of VIX, S&P/BMV IPC and Bovespa fall sharply. This establishes that the empirical representation depends materially on the declared candidate universe. It does not establish causal direction: a broad equity index is a contemporaneous market-price aggregate and includes constituents from the target market. Full window-level results are reported in Appendix D.

8.3 Out-of-sample covariance performance

For training lengths $T = 252/504$ and 126-day test blocks, the point-in-time covariance design produces 23/21 test folds. Mean out-of-sample annualized GMV volatilities are 16.0/13.5% for the sample covariance, 14.1/15.0% for PCA-k, 13.8/14.0% for PCA-$(k+2)$, 13.1/12.7% for Ledoit–Wolf, 13.5/14.2% for the structured estimator and 16.9/16.1% for $1/N$. Figure 6 compares the estimators across the two training lengths.

The structured estimator is therefore competitive in the shorter training design and improves on

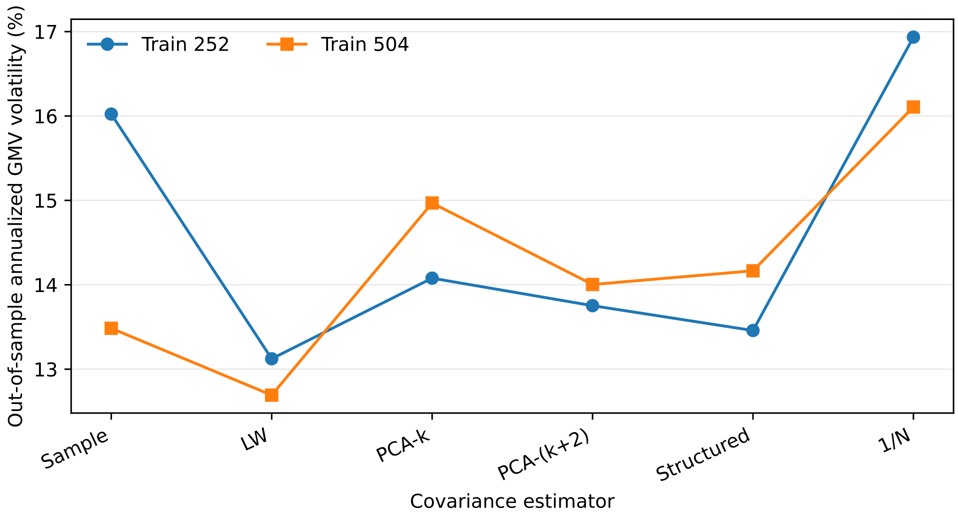

Figure 6: Point-in-time out-of-sample GMV volatility in the covariance study. Driver eligibility at each fold uses only representation selections completed before that fold. Ledoit–Wolf has the lowest mean realized volatility in both training designs; the structured estimator is closest in the shorter window.

the sample covariance and both matched-factor alternatives there. With 504 training observations, Ledoit–Wolf and the sample covariance have lower realized volatility. Average gross exposure is lower for the structured portfolios than for the shrinkage portfolios, about 1.90/1.98 versus 3.77/3.82 for $T = 252/504$, but no transaction-cost claim is inferred from this comparison. The structured covariance also understates realized variance materially: the median realized-to-predicted variance ratio is 6.45 at $T = 252$ and 4.43 at $T = 504$ (Table 8). This is consistent with the misspecification analysis rather than with exact empirical screening.

9 Conclusion

This paper develops causal separation as a framework for making the conditioning representation an object of the portfolio model rather than a fixed input. Its probabilistic component, exact screening-off, yields an exact conditional-risk decomposition: after horizon-closed conditioning, residual covariance is diagonal, while decision-time systematic risk is the uncertainty in the corresponding conditional means. A finite-dimensional innovation projection then separates represented systematic covariance from a positive-semidefinite response residual, and the familiar diagonal-plus-low-rank form is exact under affine response rather than under screening alone.

The causal interpretation is separate from the portfolio theorem. A common-cause structural partition provides sufficient conditions for screening, but observationally equivalent representations generate the same portfolio-relevant conditional information even when their causal meanings differ. The intervention experiment makes that distinction explicit: representations that are indistinguishable observationally can diverge once the measurement mechanism of a proxy is severed from the underlying state.

Once conditional moments and economically chosen sensitivity directions are supplied, portfolio choice reduces to standard constrained mean–variance analysis. The additional contribution is the way screening determines the conditional risk object and the way misspecification is propagated through that optimizer. The covariance error separates into omitted systematic response risk and residual screening error, and the exact perturbation identities show how each component changes weights and the feasible frontier.

The controlled experiments confirm the theoretical decomposition, the two information windows, the finite perturbation identities and the intervention distinction in settings where the population structure is known. The empirical applications address a different question: how well particular observable representations screen real asset panels. In the empirical screening study (Section 8.2), selected drivers materially reduce residual covariance, and the broad-market augmentation shows that the result depends on the information available to the selector. In the point-in-time covariance study (Section 8.3), the structured estimator is competitive in the shorter estimation window and shrinkage remains a strong benchmark. These findings motivate richer candidate universes and improved screening diagnostics rather than altering the theoretical results.

Further work can strengthen the empirical layer through conditional-independence diagnostics beyond pairwise residual covariance, statistically calibrated upper bounds, and dynamic or inequality-constrained portfolio problems. The main conclusion is unchanged: conditional portfolio optimization depends not only on how moments are estimated, but also on the information on which those moments are conditioned. Screening-off provides a direct way to make that information choice explicit and to quantify the portfolio consequences when the selected representation is imperfect.

Disclosure statement

The author declares that he has no relevant or material financial interests that relate to the research described in this paper.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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A Proofs and technical details

This appendix records proofs and secondary derivations supporting the results stated in the main text.

A.1 Proof of causal soundness

Proof of Proposition 3.4. Condition on the realized common path $C_{t,h}$. By Assumption 3.2, each asset path is a measurable function of $(C_{t,h}, U^i)$ and the private exogenous blocks $U^1, \ldots, U^n$ are mutually independent. Hence the pairs consisting of each asset's pre-t history and horizon return are mutually conditionally independent given $C_{t,h}$. Conditioning further on the observed pre-t asset histories preserves the product structure because, given $C_{t,h}$, those histories are coordinatewise functions of the independent private blocks. Therefore

$$\mathcal{L}(r_{t+h} \mid \mathcal{F}_t^{\mathcal{A}} \vee \sigma(C_{t,h})) = \bigotimes_{i=1}^{n} \mathcal{L}(r_{t+h}^i \mid \mathcal{F}_t^{\mathcal{A}} \vee \sigma(C_{t,h})),$$

which is Definition 3.1. The same argument does not apply automatically after adding arbitrary extra conditioning variables, hence the restriction in the statement. □

For Proposition 3.5, equality of generated sigma-algebras implies equality of conditional distributions up to almost-sure versions, so all conditional-law functionals coincide. Under a modular intervention outside the ancestral system of (C, r), the structural assignments generating (C, r) are unchanged. If the conditioning information is also unchanged, the truncated factorization yields the same conditional law.

A.2 Risk decomposition and local nonlinear response

Proof of Theorem 4.1. Since $\mathcal{G}_t \subseteq \mathcal{H}_{t,h}$, the conditional law of total covariance gives

$$\operatorname{Cov}(r_{t+h} \mid \mathcal{G}_t) = \mathbb{E}[\operatorname{Cov}(r_{t+h} \mid \mathcal{H}_{t,h}) \mid \mathcal{G}_t] + \operatorname{Cov}(\mathbb{E}[r_{t+h} \mid \mathcal{H}_{t,h}] \mid \mathcal{G}_t).$$

Under screening, the first conditional covariance is diagonal almost surely, so its conditional expectation is diagonal. The second term is a covariance matrix and is therefore positive semidefinite. $\square$

Proof of Theorem 4.2. The coefficient L in (10) is the conditional L^2 projection coefficient of $m - \mu$ on $\widetilde{\xi}$. Hence $\operatorname{Cov}(u, \widetilde{\xi} \mid \mathcal{G}_t) = 0$. Writing $m - \mu = L\widetilde{\xi} + u$ and taking conditional covariances removes the cross terms and gives

$$\operatorname{Cov}(m \mid \mathcal{G}_t) = L\Lambda L^\top + \operatorname{Cov}(u \mid \mathcal{G}_t).$$

Combining this identity with Theorem 4.1 proves (12). $\square$

Corollary A.1 (Affine response and exact low rank)**.** *Under Theorem 4.2, the following are equivalent:*

1. $\Omega_{t,h} = 0$ *almost surely;*
2. $m_{t,h} - \mu_{t,h} = L_{t,h}\widetilde{\xi}_{t,h}$ *almost surely.*

When these conditions hold,

$$Q_{t,h} = D_{t,h} + L_{t,h}\Lambda_{t,h}L_{t,h}^\top, \qquad \operatorname{rank}(Q_{t,h} - D_{t,h}) \le k. \tag{47}$$

Proof. Condition (ii) implies $u_{t,h} = 0$ and hence $\Omega_{t,h} = 0$. Conversely, $\Omega_{t,h} = \mathbb{E}[u_{t,h}u_{t,h}^\top \mid \mathcal{G}_t] = 0$ implies $u_{t,h} = 0$ almost surely. The rank bound follows because $L_{t,h}\Lambda_{t,h}L_{t,h}^\top$ has rank at most k. $\square$

Proposition A.2 (Local nonlinear response)**.** *Let $\tau \downarrow 0$ scale the horizon innovations as*

$$\xi_{t,h}^{(\tau)} = \tau z_{t,h}, \qquad \mathbb{E}[z_{t,h} \mid \mathcal{G}_t] = 0, \qquad \operatorname{Cov}(z_{t,h} \mid \mathcal{G}_t) = \Lambda_{t,h} \succ 0.$$

Suppose the horizon-closed conditional mean satisfies

$$m_{t,h}^{(\tau)} = \mu_{t,h}^{(\tau)} + \tau J_{t,h} z_{t,h} + R_{t,h}^{(\tau)}, \tag{48}$$

where $\mathbb{E}[R_{t,h}^{(\tau)} \mid \mathcal{G}_t] = 0$ and

$$\Big(\mathbb{E}[\|R_{t,h}^{(\tau)}\|_2^2 \mid \mathcal{G}_t]\Big)^{1/2} = o(\tau). \tag{49}$$

Then, in conditional operator norm,

$$\operatorname{Cov}(m_{t,h}^{(\tau)} \mid \mathcal{G}_t) = \tau^2 J_{t,h}\Lambda_{t,h}J_{t,h}^\top + o(\tau^2), \tag{50}$$

$$L_{t,h}^{(\tau)} = J_{t,h} + o(1), \qquad \Omega_{t,h}^{(\tau)} = o(\tau^2). \tag{51}$$

Under exact screening,

$$Q_{t,h}^{(\tau)} = D_{t,h}^{(\tau)} + \tau^2 J_{t,h}\Lambda_{t,h}J_{t,h}^\top + o(\tau^2). \tag{52}$$

Proof. From (48), the conditional covariance contains the leading term $\tau^2 J\Lambda J^\top$, two cross terms involving $R^{(\tau)}$, and $\mathrm{Cov}(R^{(\tau)} \mid \mathcal{G}_t)$. Conditional Cauchy–Schwarz and (49) make the cross terms and the remainder covariance $o(\tau^2)$, proving (50). Moreover,

$$L^{(\tau)} = \mathrm{Cov}(m^{(\tau)}, \tau z \mid \mathcal{G}_t)(\tau^2\Lambda)^{-1} = J + o(1).$$

The conditional linear projection minimizes residual second moment, so its residual covariance is $o(\tau^2)$, giving (51). The final statement follows from Theorem 4.1. □

Proposition A.3 (Coordinate invariance of sensitivity constraints)**.** *Let $z' = \psi(z)$ be a C^1 local diffeomorphism. Transport the neutral state directions by $U'_t = D\psi(Z_t)U_t$ and write the same conditional-mean map in the new coordinates, with Jacobian S'_t. Then*

$$S'_t = S_t D\psi(Z_t)^{-1}, \qquad U_t'^\top S_t'^\top = U_t^\top S_t^\top = \Gamma_t. \tag{53}$$

Hence the admissible portfolio subspace does not depend on the local coordinate chart used for the selected decision state.

Proof. The first identity is the chain rule applied to $g_t \circ \psi^{-1}$. Substitution into the definition of Γ_t gives the second identity. □

A.3 Projected Markowitz derivation

Proof of Theorem 5.1. The Lagrangian for (18) is

$$\mathcal{L}(w, \lambda) = \gamma w^\top \mu - \frac{1}{2} w^\top Q w + \lambda^\top (b - Aw).$$

The first-order conditions give

$$Qw = \gamma\mu - A^\top\lambda, \qquad Aw = b.$$

Substituting $w = \gamma Q^{-1}\mu - Q^{-1}A^\top\lambda$ into the constraint gives

$$\lambda = (AQ^{-1}A^\top)^{-1}(\gamma A Q^{-1}\mu - b).$$

Substitution yields $w^\star = w_0 + \gamma M\mu$ with (20)–(21). If N spans $\ker A$, the feasible displacement from any particular feasible point is Nv. Minimizing risk on that tangent space gives the reduced inverse $(N^\top QN)^{-1}$ and therefore (22). Positive definiteness of Q gives uniqueness. □

The frontier identities follow from $Aw_0 = b$, $AM = 0$ and $MQM = M$. In particular, $w_0^\top QM = 0$ and

$$(w_0 + \gamma M\mu)^\top Q(w_0 + \gamma M\mu) = \sigma_0^2 + \gamma^2 \mu^\top M\mu.$$

Equation (26) is the Rayleigh-quotient representation on the feasible tangent space.

Proposition A.4 (Residual-risk floor and mean stability)**.** *If $D \succeq \underline{\sigma}^2 I$ with $\underline{\sigma}^2 > 0$, then*

$$Q \succeq \underline{\sigma}^2 I, \qquad \|Q^{-1}\|_2 \le \underline{\sigma}^{-2}, \qquad \|M\|_2 \le \underline{\sigma}^{-2},$$

and, holding (Q, A, b) fixed,

$$\|w^\star(\mu + \delta\mu) - w^\star(\mu)\|_2 \le \gamma\underline{\sigma}^{-2}\|\delta\mu\|_2.$$

Proof. The risk decomposition gives $Q = D + \Sigma^{\text{sys}} \succeq D$. Also $M = Q^{-1/2} P Q^{-1/2}$ for the orthogonal projector P onto $\ker(AQ^{-1/2})$, so $\|M\|_2 \le \|Q^{-1}\|_2$. The last inequality follows from $w^\star(\mu + \delta\mu) - w^\star(\mu) = \gamma M \delta\mu$. $\square$

Proposition A.5 (Structured solve at fixed dimension)**.** *Suppose the affine representation is exact, $Q = D + L\Lambda L^\top$, with D diagonal positive definite, $L \in \mathbb{R}^{n\times k}$ and p equality constraints in $Aw = b$. Then the projected Markowitz solution can be computed without forming a dense $n \times n$ inverse, with arithmetic cost*

$$O\big(n(k+p)^2 + (k+p)^3\big) \tag{54}$$

and working memory $O(n(k+p))$.

Proof. Factor the $k\times k$ matrix $\Lambda^{-1} + L^\top D^{-1} L$ once. Woodbury then gives $Q^{-1}\mu$ and the p columns of $Q^{-1}A^\top$. Forming the required cross-products costs $O(n(k+p)^2)$; the remaining dense factorizations involve matrices of dimension at most $k+p$. The stated memory bound follows from storing L, $A^\top$ and the corresponding transformed columns. $\square$

A.4 Finite covariance perturbations and diagnostic bound

Proof of Theorem 6.1. Let N span $\ker A$. By (22),

$$M = N(N^\top \bar{Q} N)^{-1} N^\top, \qquad M_R = N\{N^\top(\bar{Q} + R)N\}^{-1} N^\top.$$

The resolvent identity for the two reduced inverse matrices gives

$$M_R - M = -MRM_R = -M_R R M.$$

For the portfolio weights, subtract the two KKT systems. Since $d := w_R^\star - w^\star \in \ker A$, multiplication by M_R gives $d = -M_R R w^\star$. Reversing the roles of $\bar{Q}$ and $\bar{Q}_R$ gives $d = -MRw_R^\star$. Finally,

$$\Delta_R - \Delta = \mu^\top (M_R - M)\mu = -(M\mu)^\top R (M_R \mu).$$

$\square$

If $\|MR\|_2 < 1$, the identity $d = -MR(w^\star + d)$ yields

$$\|d\|_2 \le \frac{\|MR\|_2}{1 - \|MR\|_2} \|w^\star\|_2.$$

Under the floor (27), $\|M\|_2 \le \underline{\sigma}^{-2}$ and Weyl's inequality gives $\bar{Q} + R \succeq (\underline{\sigma}^2 - \|R\|_2)I$, which yields (39)–(40). Setting $\bar{Q} = Q^0$ and $R = \Omega \succeq 0$, the reduced risk matrix increases in Loewner order and its inverse decreases, proving (41).

Proof of Proposition 6.2. Because $E_{ii} = 0$,

$$\|E\|_F^2 = \sum_{i\neq j} E_{ij}^2 \le (\epsilon^{\text{cov}})^2 \sum_{i \neq j} s_i^2 s_j^2 = (\epsilon^{\text{cov}})^2 \left\{ \left(\sum_i s_i^2 \right)^2 - \sum_i s_i^4 \right\}.$$

Use $\|E\|_2 \le \|E\|_F$ and take square roots. $\square$

Corollary A.6 (Joint local mean and covariance perturbation). *Let $\bar{Q}_\tau = \bar{Q} + \tau R$ and $\mu_\tau = \mu + \tau a$, with (A, b, γ) fixed. At $\tau = 0$,*

$$\left.\frac{dw^\star_\tau}{d\tau}\right|_{\tau=0} = \gamma M a - M R w^\star, \tag{55}$$

$$\left.\frac{d\Delta_\tau}{d\tau}\right|_{\tau=0} = 2a^\top M\mu - (M\mu)^\top R(M\mu). \tag{56}$$

Thus mean and covariance error enter through distinct first-order channels.

Proof. Differentiate $w^\star_\tau = w_{0,\tau} + \gamma M_\tau \mu_\tau$ through the KKT system, or equivalently use the covariance derivative implied by Theorem 6.1. The mean perturbation contributes $\gamma M a$ and the covariance perturbation contributes $-MRw^\star$. Differentiating $\Delta_\tau = \mu_\tau^\top M_\tau \mu_\tau$ gives the second identity. □

B Additional controlled results

B.1 Full covariance-estimation grid

Table 2 reports the complete covariance experiment rather than only the $T = 252$ slice displayed in Figure 2. It includes under- and over-specified PCA dimensions to show that much of the gap to the structured estimator disappears when PCA is supplied with the correct factor dimension.

Table 2: Controlled covariance experiment: out-of-sample annualized GMV volatility and variance calibration. There are 250 Monte Carlo replications per cell and 252 out-of-sample observations.

n	T	Sample	LW	PCA-(m−2)	PCA-m	PCA-(m+2)	Structured	Oracle
50	126	3.91	3.44	4.30	3.15	3.20	3.15	3.05
50	252	3.42	3.31	4.30	3.11	3.15	3.10	3.05
50	504	3.24	3.22	4.33	3.11	3.15	3.10	3.08
100	126	4.68	2.55	3.04	2.15	2.16	2.15	2.08
100	252	2.68	2.40	3.03	2.10	2.11	2.10	2.07
100	504	2.31	2.26	2.99	2.09	2.09	2.08	2.07
200	126	276.90	1.87	2.16	1.48	1.49	1.49	1.44
200	252	3.21	1.87	2.13	1.46	1.46	1.46	1.44
200	504	1.85	1.71	2.08	1.45	1.45	1.45	1.44
Calibration: realized/predicted variance (1 = perfect)								
50	252	1.57	1.10	1.12	1.13	1.20	1.04	1.00
100	252	2.76	1.44	1.12	1.08	1.12	1.04	0.99
200	252	25.52	2.43	1.10	1.07	1.09	1.04	1.00

B.2 Finite-perturbation calibration

The response-error design uses

$$m_\rho = \mu + B\eta + \sqrt{\frac{\rho\underline{\sigma}^2}{2}}\, c \left[\left(\frac{\eta_1}{\sqrt{\Lambda_{11}}}\right)^2 - 1\right],$$

where c is aligned with the baseline admissible speculative direction. Since the centered squared Gaussian term has variance two and is orthogonal to the linear innovation span,

$$\Omega_\rho = \rho\underline{\sigma}^2 cc^\top, \qquad E = 0.$$

For the screening-error design, $\Omega = 0$ and a two-asset zero-diagonal residual block is scaled to have spectral norm $\rho\underline{\sigma}^2$. Using both signs shows that E has no one-sided effect on Δ.

C Empirical design details and declared universes

C.1 Twelve-stock panel and declared driver universe

Table 3 lists the 12 equities in the screening panel. Tables 4–5 list all 127 source driver labels and records the transformation or exclusion applied before selection. Of the 127 source series, four are excluded ex ante: the monthly S&P CoreLogic Case–Shiller series, weekly initial jobless claims, the CSI 300 series with sentinel history, and the illiquid USD–CLP risk reversal. Of the remaining series, 21 are first-differenced and 102 are log-differenced, leaving the 123 daily candidates used in the rolling exercise.

Table 3: Twelve equities in the screening panel, using the source-workbook labels.

APPLE INC	BLACKROCK INC	DOVER CORP
FORD MOTOR CO	GENERAL ELECTRIC CO	GOLDMAN SACHS GROUP INC
MCDONALD'S CORP	NVIDIA CORP	ORACLE CORP
PACKAGING CORP OF AMERICA	PFIZER INC	SCHLUMBERGER LTD

The preprocessing is fixed before the rolling selection. Asset and driver dates are intersected, driver levels may be carried forward for at most two observations, and rows on which all asset levels are stale are removed before asset log returns are formed. Series identified as rates or yields are first-differenced; a remaining series with non-positive observations is also first-differenced because a log change is undefined. Other included series are log-differenced. After alignment to the asset-return dates, remaining transformed driver missings are set to zero and series with nonzero changes on at most half of the dates would be removed; no additional driver is lost at this final filter in the present panel.

Table 4: Declared driver universe, part I. Labels reproduce the source workbook; the treatment column records the transformation or exclusion applied before rolling selection.

Series	Treatment	Series	Treatment
1-3 Year EU	Log diff.	EUR SWAP ANN (VS 6M) 30Y	First diff.
3-5 Years EU	Log diff.	EUR SWAP ANN (VS 6M) 5Y	First diff.
7-10 Years EU	Log diff.	EUR-AUD X-RATE	Log diff.
AUD-CAD X-RATE	Log diff.	EUR-CAD X-RATE	Log diff.
AUD-JPY X-RATE	Log diff.	EUR-CHF X-RATE	Log diff.
AUD-NZD X-RATE	Log diff.	EUR-CNY X-RATE	Log diff.
Argentine Peso Spot	Log diff.	EUR-CZK X-RATE	Log diff.
Australian Dollar Spot	Log diff.	EUR-GBP X-RATE	Log diff.
BBG Commodity	Log diff.	EUR-JPY X-RATE	Log diff.
BBG Euro Ser-E Gov 1-3 Yr	Log diff.	EUR-NOK X-RATE	Log diff.
BEIG1T	Log diff.	EUR-PLN X-RATE	Log diff.
BONOS Y OBLIG DEL ESTADO	First diff.	EUR-SEK X-RATE	Log diff.
BRAZIL IBOVESPA INDEX	Log diff.	Eonia Capitalization Index 7 D	First diff.
BUONI POLIENNALI DEL TES	First diff.	Euro Spot	Log diff.
Brazilian Real Spot	Log diff.	Euro-Aggregate	Log diff.
British Pound Spot	Log diff.	FRANCE (GOVT OF)	First diff.
CAC 40 INDEX	Log diff.	FTSE 100 INDEX	Log diff.
CAD-CHF X-RATE	Log diff.	FTSE MIB INDEX	Log diff.
CAD-JPY X-RATE	Log diff.	GBP-AUD X-RATE	Log diff.
CHF-JPY X-RATE	Log diff.	GBP-CAD X-RATE	Log diff.
CSI 300 INDEX	Excluded (sentinel)	GERMANY GOVT BND 10 YR DBR	First diff.
Canadian Dollar Spot	Log diff.	GERMANY GOVT BND 2 YR BKO	First diff.
Cboe Volatility Index	Log diff.	GERMANY GOVT BND 5 YR OBL	First diff.
Corporate	Log diff.	Generic 1st 'C ' Future	Log diff.
Credit	Log diff.	Generic 1st 'CF' Future	Log diff.
DAX INDEX	Log diff.	Generic 1st 'CO' Future	Log diff.
DB Euro Overnight Rate	Log diff.	Generic 1st 'CT' Future	First diff.
DOLLAR INDEX SPOT	Log diff.	Generic 1st 'FN' Future	Log diff.
EM USD Aggregate	Log diff.	Generic 1st 'FV' Future	Log diff.
EUR SWAP ANN (VS 6M) 10Y	First diff.	Generic 1st 'GX' Future	Log diff.
EUR SWAP ANN (VS 6M) 1Y	First diff.	Generic 1st 'HG' Future	Log diff.
EUR SWAP ANN (VS 6M) 2Y	First diff.	Generic 1st 'HO' Future	Log diff.

Table 5: Declared driver universe, part II. Together with Table 4, this lists all 127 source driver series.

Series	Treatment	Series	Treatment
Generic 1st 'KC' Future	Log diff.	Polish Zloty Spot	Log diff.
Generic 1st 'LC' Future	Log diff.	RF Global Inflation-Linked	Log diff.
Generic 1st 'NG' Future	Log diff.	RF Global Treasury	Log diff.
Generic 1st 'OE' Future	Log diff.	Russian Ruble SPOT (TOM)	First diff.
Generic 1st 'QS' Future	Log diff.	S&P CoreLogic Case-Shiller 20-	Excluded (monthly)
Generic 1st 'RX' Future	Log diff.	S&P/BMV IPC	Log diff.
Generic 1st 'S ' Future	First diff.	S. African Rand Spot	Log diff.
Generic 1st 'SB' Future	Log diff.	Silver Spot \$/Oz	Log diff.
Generic 1st 'TY' Future	Log diff.	Singapore Dollar Spot	Log diff.
Generic 1st 'US' Future	Log diff.	South Korean Won Spot	Log diff.
Generic 1st 'W ' Future	Log diff.	Swedish Krona Spot	Log diff.
Generic 1st 'Z ' Future	Log diff.	Swiss Franc Spot	Log diff.
Global Aggregate	Log diff.	TR/CC CRB ER Index	Log diff.
Global High Yield	Log diff.	Treasury	Log diff.
Gold Spot \$/Oz	Log diff.	Turkish Lira Spot	Log diff.
Government-Related	Log diff.	U.S. Aggregate	Log diff.
HANG SENG INDEX	Log diff.	U.S. Corporate High Yield	Log diff.
IBEX 35 INDEX	Log diff.	U.S. TIPS	Log diff.
INDIA GOVERNMENT BOND	First diff.	U.S. Treasury	Log diff.
Indian Rupee Spot	Log diff.	UK Gilts 10 Yr	Log diff.
Japanese Yen Spot	Log diff.	UK Gilts 30 Year	Log diff.
LME ALUMINUM 3MO (\$)	Log diff.	UK Gilts 5 Year	First diff.
LME COPPER 3MO (\$)	Log diff.	US Generic Govt 10 Yr	First diff.
LME COPPER SPOT (\$)	Log diff.	US Generic Govt 2 Yr	First diff.
MSCI INDIA	Log diff.	US Generic Govt 5 Yr	First diff.
Mexican Peso Spot	Log diff.	US Initial Jobless Claims SA	Excluded (weekly)
NETHERLANDS GOVERNMENT	First diff.	USD-CAD X-RATE	Log diff.
NIKKEI 225	Log diff.	USD-CHF X-RATE	Log diff.
NOK-SEK X-RATE	Log diff.	USD-CLP RR 25D 3M	Excluded (illiquid)
New Zealand Dollar Spot	Log diff.	USD-GBP X-RATE	Log diff.
Norwegian Krone Spot	Log diff.	USD-JPY X-RATE	Log diff.
Pan-European High Yield	Log diff.		

C.2 Rolling representation-selection rule

For a candidate set $\mathcal{D}$, let

$$J(\mathcal{D}) := \widehat{\epsilon}^{\text{cov}}_{\text{train}}(\mathcal{D}) + \kappa|\mathcal{D}|, \qquad \kappa = 0.01, \qquad |\mathcal{D}| \leq 8.$$

The reported selector is a deterministic local search for a low value of J, not a claim of global minimization over all 2^{123} subsets:

1. Start from $\mathcal{D} = \varnothing$ and compute the unconditioned training diagnostic.
2. Among all one-driver additions, choose the driver giving the smallest residual-correlation diagnostic. Add it only if the decrease in $\widehat{\epsilon}^{\text{cov}}_{\text{train}}$ is at least κ; otherwise stop the forward pass. Repeat until no improving addition exists or eight drivers have been selected.
3. Starting from the selected set, scan the selected drivers and remove the first driver whose deletion raises the diagnostic by no more than κ. Restart the scan after each deletion and stop when no deletion weakly improves J.
4. Candidate ties are resolved by the fixed source-column order. The selected labels are then frozen for the disjoint holdout diagnostics described in the main text.

This procedure is equivalent to greedy coordinate moves on J because every forward candidate has the same one-unit complexity increment and every backward candidate the same one-unit decrement.

C.3 Omitted-driver floor used in the VIX stress

For the VIX-removal diagnostic, let ξ be the VIX innovation after projecting it on an intercept and the innovations of the re-selected replacement set. Let b be the vector of coefficients obtained by regressing the replacement-set return residuals on $(1, \xi)$, and let s_i^2 denote their residual variances. The fitted omitted-channel correlation matrix is

$$\widehat{C}_{ij}^{\text{omit}} = \frac{b_i b_j \, \widehat{\text{Var}}(\xi)}{s_i s_j}, \qquad i \neq j, \tag{57}$$

with zero diagonal. The reported model-based floor is $\max_{i \neq j} |\widehat{C}_{ij}^{\text{omit}}|$. It is a directional omitted-channel diagnostic; it is not a lower bound on the realized maximum residual-correlation statistic after re-selection.

C.4 Fixed equity universe for the covariance study

Table 6 lists the fixed survivorship-conditioned ticker set used in the point-in-time covariance comparison. The same names are used for every estimator and every fold.

Table 6: Fixed ticker universe used in the covariance comparison.

A	AAP	AAPL	ABT	ACGL	ACN
ADBE	ADI	ADM	ADP	ADSK	AEE
AEP	AES	AFL	AIG	AIT	AIV
AIZ	AJG	AKAM	ALB	ALGN	ALK
ALL	AMAT	AMD	AME	AMG	AMGN
AMP	AMT	AMZN	AN	ANF	ANSS
AON	AOS	APA	APD	APH	ARC
ARE	ASH	ATGE	ATI	ATO	AVB
AVGO	AVY	AWK	AXON	AXP	AYI
AZO	BA	BAC	BALL	BAX	BBWI
BBY	BC	BCO	BDX	BEN	BG
BIIB	BIO	BK	BKNG	BLDR	BLK
BMY	BR	BRO	BSX	BUD	BWA
BX	BXP	C	CAG	CAH	CAL
CAR	CAT	CB	CBOE	CBRE	CCI
CCK	CCL	CCU	CDNS	CE	CF
CHD	CHRW	CHTR	CI	CIEN	CINF
CL	CLF	CLX	CMA	CMCSA	CME
CMG	CMI	CMS	CNC	CNP	CNX
COF	COO	COP	COR	COST	CPB
CPRT	CPT	CRL	CRM	CSCO	CSGP
CSR	CSX	CTAS	CTL	CTRA	CTSH
CVS	CVX	D	DAL	DD	DDS
DE	DFS	DG	DGX	DHI	DHR
DIS	DISH	DLR	DLTR	DLX	DOV

D Additional empirical results

D.1 Candidate-universe sensitivity

Table 7 summarizes the broad U.S. market augmentation using the CRSP value-weighted U.S. market return (including distributions, `vwretd`) as the proxy. The re-selection column repeats the original rolling protocol after adding the market proxy to the candidate universe. The final column keeps each original selected set fixed and adds the market proxy mechanically. The latter is a pure incremental-information diagnostic and is not a competing selection rule. Table 7 reports the aggregate diagnostics, while Figure 7 shows the window-by-window change and the selection-frequency substitution.

Table 7: Candidate-universe sensitivity of the empirical screening study. Percentages are descriptive shares across overlapping rolling windows.

Quantity	Original 123	Re-select + market	Add market only
Median set-refit max $\|\rho\|$	0.384	0.287	0.283
Median frozen-model max $\|\rho\|$	0.450	0.305	0.285
Windows with lower set-refit diagnostic	–	87%	95%
Windows with lower frozen diagnostic	–	90%	95%

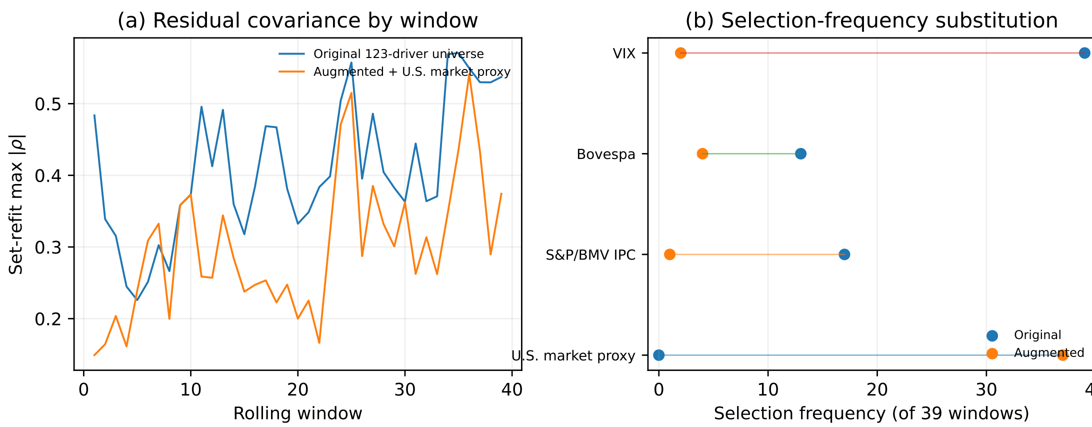

Figure 7: Candidate-universe sensitivity by rolling window. Panel (a) compares the original and augmented set-refit residual-correlation diagnostic. Panel (b) shows the change in selection frequencies. The result is observational and does not assign causal status to the market index.

The market proxy enters 37 of 39 augmented selections. The VIX frequency falls from 39/39 to 2/39, S&P/BMV IPC from 17/39 to 1/39, and Bovespa from 13/39 to 4/39. The result is consistent with those variables carrying information that partly substitutes for broad U.S. market information in the compact universe. It does not show that the market index is the unique or structural common cause.

D.2 Point-in-time covariance comparison

Table 8 reports the main statistics underlying Figure 6. The comparison is conditional on the fixed survivorship-conditioned equity panel defined in Appendix C and should not be interpreted as an investable-universe backtest.

Table 8: Point-in-time covariance comparison. Realized volatility and gross exposure are means across 126-day test blocks. “Calib.” is the median realized-to-predicted variance ratio; one denotes exact variance calibration.

	Training $T = 252$			Training $T = 504$		
Estimator	Vol. (%)	Gross	Calib.	Vol. (%)	Gross	Calib.
Sample	16.02	6.67	8.94	13.48	4.79	2.63
Ledoit–Wolf	13.12	3.77	3.12	12.69	3.82	2.05
PCA-k	14.08	2.20	8.93	14.97	2.26	8.61
PCA-$(k+2)$	13.75	2.60	4.80	14.00	2.80	3.75
Structured	13.46	1.90	6.45	14.17	1.98	4.43
$1/N$	16.93	1.00	–	16.11	1.00	–

With the shorter training window, the structured covariance reduces realized GMV volatility relative to the sample covariance and the matched-factor alternatives while remaining close to Ledoit–Wolf. The calibration columns show that all optimized covariance estimators underpredict realized variance in these folds, with the structured estimator’s median ratio falling from 6.45 to 4.43 as the training window lengthens. With 504 training observations, both Ledoit–Wolf and the sample covariance have lower realized volatility than the structured estimator. Reporting both regimes makes the comparison independent of a single favorable estimation window.